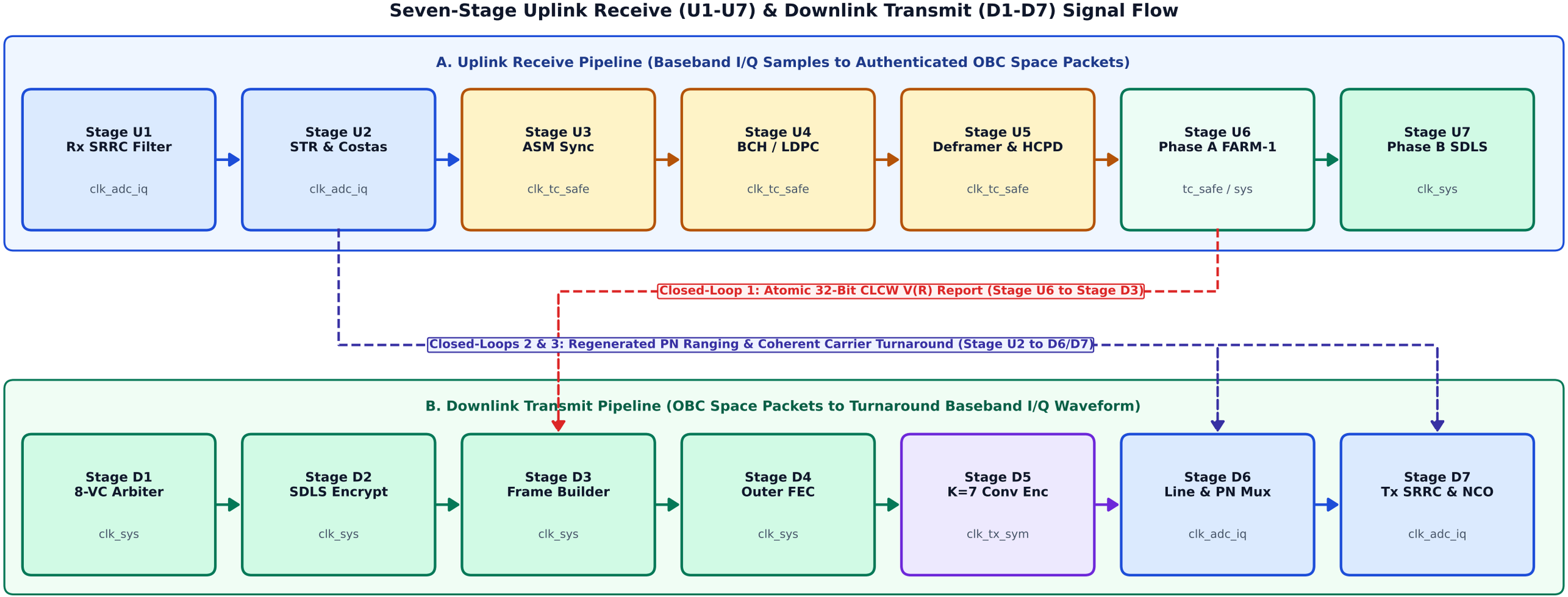
Seven-Stage Uplink Receive (U1-U7) & Downlink Transmit (D1-D7) Signal Flow
A. Uplink Receive Pipeline (Baseband I/Q Samples to Authenticated OBC Space Packets)
Stage U1
Rx SRRC Filter
clk_adc_iq
Stage U2
STR & Costas
clk_adc_iq
Stage U3
ASM Sync
clk_tc_safe
Stage U4
BCH / LDPC
clk_tc_safe
Stage U5
Deframer & HCPD
clk_tc_safe
Stage U6
Phase A FARM-1
tc_safe / sys
Stage U7
Phase B SDLS
clk_sys
Closed-Loop 1: Atomic 32-Bit CLCW V(R) Report (Stage U6 to Stage D3)
Closed-Loops 2 & 3: Regenerated PN Ranging & Coherent Carrier Turnaround (Stage U2 to D6/D7)
B. Downlink Transmit Pipeline (OBC Space Packets to Turnaround Baseband I/Q Waveform)
Stage D1
8-VC Arbiter
clk_sys
Stage D2
SDLS Encrypt
clk_sys
Stage D3
Frame Builder
clk_sys
Stage D4
Outer FEC
clk_sys
Stage D5
K=7 Conv Enc
clk_tx_sym
Stage D6
Line & PN Mux
clk_adc_iq
Stage D7
Tx SRRC & NCO
clk_adc_iq